

Boss Challenge — Paper 04 (Class 7)

Physical & Chemical Changes | Weather, Climate & Adaptations | Lines & Angles

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. A change in which NO new substance is formed and which can often be reversed is a:
(A) nuclear change
(B) physical change
(C) permanent change always
(D) chemical change
2. The day-to-day condition of the atmosphere (temperature, rain, wind) at a place is called:
(A) weather
(B) season
(C) humidity
(D) climate
3. Two angles whose measures add up to 90° are called:
(A) reflex angles
(B) vertically opposite
(C) complementary angles
(D) supplementary angles
4. Burning a piece of paper is a:
(A) physical change
(B) chemical change
(C) reversible change
(D) change of state only
5. The average of weather conditions over many years is called the:
(A) climate
(B) forecast
(C) weather
(D) temperature
6. Two angles that add up to 180° are called:
(A) complementary angles
(B) equal angles
(C) acute angles
(D) supplementary angles
7. Rusting of iron is a chemical change that needs both:
(A) oxygen and moisture (water)
(B) only oxygen
(C) only heat
(D) only water

8. Humidity in the air is a measure of the amount of:
- (A) water vapour
 - (B) dust
 - (C) heat
 - (D) oxygen
9. An angle that is exactly 90° is called a:
- (A) right angle
 - (B) acute angle
 - (C) straight angle
 - (D) obtuse angle
10. Which of these is a physical change?
- (A) burning of wood
 - (B) rusting of iron
 - (C) souring of milk
 - (D) melting of ice
11. The main source of energy that drives the weather is the:
- (A) wind
 - (B) rivers
 - (C) Sun
 - (D) Moon
12. If one angle of a linear pair is 70° , the other angle is:
- (A) 110°
 - (B) 70°
 - (C) 20°
 - (D) 290°
13. Souring (curdling) of milk is a chemical change because:
- (A) the milk just gets colder
 - (B) it can be reversed by heating
 - (C) a new substance (curd) is formed that can't be turned back into milk
 - (D) only its shape changes
14. A polar bear has a thick layer of fat and dense fur mainly to:
- (A) keep warm in the freezing cold
 - (B) swim faster
 - (C) look white
 - (D) find food
15. The complement of a 30° angle is:
- (A) 60°
 - (B) 70°
 - (C) 330°
 - (D) 150°

16. Which is a sign that a chemical change has taken place?
- (A) it can be reversed easily
 - (B) it just gets warmer by touch
 - (C) the object only changes shape
 - (D) a new substance with a different colour/smell forms
17. Which region's climate is hot and dry, with sandy adaptations like the camel?
- (A) rainforest
 - (B) polar region
 - (C) a desert
 - (D) ocean
18. Vertically opposite angles (formed when two lines cross) are always:
- (A) supplementary
 - (B) complementary
 - (C) always 90°
 - (D) equal
19. Dissolving sugar in water is a physical change because:
- (A) a new substance forms
 - (B) it gives off a gas
 - (C) the sugar can be got back by evaporating the water
 - (D) it cannot be reversed
20. Birds like the Siberian crane fly thousands of kilometres to India in winter. This is called:
- (A) migration
 - (B) adaptation by fur
 - (C) hibernation
 - (D) camouflage
21. The supplement of a 110° angle is:
- (A) 20°
 - (B) 80°
 - (C) 70°
 - (D) 250°
22. Heating a magnesium ribbon makes it burn with a bright light, leaving a white ash. This is a:
- (A) reversible change
 - (B) change of state
 - (C) physical change
 - (D) chemical change
23. Penguins in the cold huddle together in large groups mainly to:
- (A) look bigger
 - (B) find food
 - (C) stay warm
 - (D) swim

24. An angle greater than 90° but less than 180° is called:
- (A) right
 - (B) reflex
 - (C) obtuse
 - (D) acute
25. Cutting a log of wood is physical, but burning the same wood is:
- (A) reversible
 - (B) also physical
 - (C) a chemical change (new substances form)
 - (D) no change
26. Which animal is adapted to a hot, wet tropical rainforest?
- (A) a penguin
 - (B) a monkey with a strong grip for trees
 - (C) a camel
 - (D) a polar bear
27. An angle less than 90° is called:
- (A) straight
 - (B) right
 - (C) acute
 - (D) obtuse
28. Which pair correctly matches the change type?
- (A) both are physical
 - (B) melting is chemical
 - (C) melting wax = physical; burning the wick = chemical
 - (D) both are chemical
29. Maximum and minimum temperatures of a day are measured using a:
- (A) barometer
 - (B) thermometer
 - (C) rain gauge
 - (D) anemometer
30. Two angles are complementary and one of them is 25° . The other is:
- (A) 65°
 - (B) 25°
 - (C) 155°
 - (D) 75°
31. Galvanisation means coating iron with a layer of ____ to prevent rusting.
- (A) paper
 - (B) gold
 - (C) salt
 - (D) zinc

32. Which statement is TRUE?
- (A) weather and climate are the same
 - (B) climate changes every hour
 - (C) weather is the 30-year average
 - (D) weather can change in a day, but climate is the long-term pattern
33. If two lines cross and one angle is 50° , the angle vertically opposite it is:
- (A) 130°
 - (B) 50°
 - (C) 90°
 - (D) 40°
34. Which change is reversible (and therefore physical)?
- (A) cooking an egg
 - (B) freezing water into ice
 - (C) burning coal
 - (D) rusting a nail
35. A camel can survive in the desert partly because it:
- (A) hibernates all summer
 - (B) can go a long time without water
 - (C) breathes through gills
 - (D) has thick white fur for snow
36. Angles on a straight line (a linear pair) always add up to:
- (A) 100°
 - (B) 90°
 - (C) 180°
 - (D) 360°
37. Photosynthesis (a plant making food) is a:
- (A) change of state
 - (B) no change
 - (C) chemical change
 - (D) physical change
38. Why do many polar animals have white fur or feathers?
- (A) to absorb heat
 - (B) to look pretty
 - (C) to blend in with snow (camouflage)
 - (D) to scare enemies
39. The angle that measures exactly 180° (a straight line) is a:
- (A) complete angle
 - (B) reflex angle
 - (C) right angle
 - (D) straight angle

40. When iron filings and sulphur powder are simply MIXED, it is a physical change; but when the mixture is HEATED strongly it becomes:

- (A) still just a mixture
- (B) a chemical change (iron sulphide forms)
- (C) a physical change
- (D) reversible

41. Which is NOT an element of weather?

- (A) the price of vegetables
- (B) rainfall
- (C) temperature
- (D) wind speed

42. If an angle is 35° , its supplement minus its complement equals:

- (A) 35°
- (B) 90°
- (C) 110°
- (D) 180°

43. Ripening of a fruit (raw mango turning sweet and yellow) is a:

- (A) physical change
- (B) reversible change
- (C) chemical change
- (D) a state change

44. Animals in tropical rainforests are often brightly coloured or very loud to:

- (A) dig burrows
- (B) keep warm
- (C) communicate or warn in the dense forest
- (D) store water

45. Adjacent angles are angles that:

- (A) are always equal
- (B) share a common vertex and a common arm
- (C) always add to 90°
- (D) are far apart

46. Which of these is the BEST example of a change that is both physical and reversible?

- (A) boiling water into steam
- (B) digesting food
- (C) burning petrol
- (D) baking a cake

47. The kind of plants and animals found in a place depends mostly on its:
- (A) flag
 - (B) one day's weather
 - (C) name
 - (D) climate
48. If a transversal cuts two PARALLEL lines, the corresponding angles are:
- (A) complementary
 - (B) supplementary
 - (C) zero
 - (D) equal
49. Which observation usually means a chemical change happened?
- (A) the object is cut smaller
 - (B) a gas (bubbles) is given off with a new smell
 - (C) it dissolves and can be recovered
 - (D) it is bent into a shape
50. Which weather instrument measures rainfall?
- (A) rain gauge
 - (B) compass
 - (C) barometer
 - (D) thermometer
51. Two angles of a linear pair are equal. Each angle is:
- (A) 45°
 - (B) 60°
 - (C) 90°
 - (D) 180°
52. Cooking raw rice into soft cooked rice is a:
- (A) no change
 - (B) change of state only
 - (C) chemical change
 - (D) physical change
53. Which adaptation helps a fish-eating polar animal see prey and stay hidden?
- (A) thin bare skin
 - (B) bright red colour
 - (C) large ears for heat loss
 - (D) white coat for camouflage in snow and ice
54. Co-interior (allied) angles between two parallel lines add up to:
- (A) 180°
 - (B) equal
 - (C) 360°
 - (D) 90°

55. Which is a physical change?
- (A) burning a candle
 - (B) breaking a glass tumbler
 - (C) curdling milk
 - (D) rusting iron
56. A desert plant like a cactus has thick stems and spines mainly to:
- (A) store water and reduce water loss
 - (B) attract rain
 - (C) float on water
 - (D) stay warm at night
57. Alternate interior angles between two parallel lines cut by a transversal are:
- (A) zero
 - (B) complementary
 - (C) equal
 - (D) supplementary
58. Magnetising an iron nail (making it a temporary magnet) is a:
- (A) chemical change
 - (B) irreversible change
 - (C) physical change
 - (D) burning
59. Which is the best example of 'weather' rather than 'climate'?
- (A) it is raining heavily right now
 - (B) this region is hot and dry all year
 - (C) summers here are always wet
 - (D) the area has a cold climate
60. The reflex angle for a 70° angle (going the long way round) is:
- (A) 70°
 - (B) 290°
 - (C) 20°
 - (D) 110°
61. Stretching a rubber band is a physical change because:
- (A) a new rubber forms
 - (B) it returns to its shape and no new substance forms
 - (C) it can't return
 - (D) it burns
62. Which is a behaviour (not a body feature) that helps animals survive harsh weather?
- (A) thick fur
 - (B) a layer of fat
 - (C) white skin
 - (D) migrating to warmer places in winter

63. An angle of 360° (one full turn) is called a:
- (A) complete angle
 - (B) right angle
 - (C) reflex angle
 - (D) straight angle
64. Which of these gives out heat AND light, showing a chemical change?
- (A) cutting paper
 - (B) burning of a candle
 - (C) melting ice
 - (D) dissolving sugar
65. In very cold regions, animals often have small ears and a rounded body to:
- (A) lose less body heat
 - (B) store food
 - (C) run faster
 - (D) hear better
66. Two angles are supplementary and one is three times the other. The smaller angle is:
- (A) 60°
 - (B) 30°
 - (C) 90°
 - (D) 45°
67. Which change can be reversed by simply cooling?
- (A) steam turning back to water
 - (B) ash turning back to wood
 - (C) rust turning back to shiny iron
 - (D) curd turning back to milk
68. The chief reason different places have different climates is differences in:
- (A) their names
 - (B) number of people
 - (C) the day's wind only
 - (D) temperature and rainfall over the year
69. If two parallel lines are cut by a transversal and one angle is 65° , its corresponding angle is:
- (A) 115°
 - (B) 65°
 - (C) 90°
 - (D) 25°
70. Which statement about physical and chemical changes is correct?
- (A) neither can be reversed
 - (B) in a physical change no new substance forms; in a chemical change a new substance forms
 - (C) both form new substances
 - (D) chemical changes never give heat

71. A thick layer of fat (blubber) under the skin of polar animals mainly works as:
- (A) to float only
 - (B) for swimming speed
 - (C) insulation against cold
 - (D) a food store only
72. The complement of an angle is 40° . The angle itself is:
- (A) 50°
 - (B) 140°
 - (C) 60°
 - (D) 40°
73. Which of these is an IRREVERSIBLE change?
- (A) melting butter
 - (B) burning of a matchstick
 - (C) folding paper
 - (D) freezing water
74. Which set correctly pairs an animal with its climate?
- (A) penguin - desert
 - (B) camel - polar region
 - (C) polar bear - cold polar region
 - (D) monkey - polar ice
75. Two angles are complementary. If one is twice the other, the larger angle is:
- (A) 120°
 - (B) 30°
 - (C) 45°
 - (D) 60°
76. Crystallisation, used to get pure sugar/salt crystals from a solution, is a:
- (A) burning
 - (B) irreversible change
 - (C) physical change
 - (D) chemical change
77. On a hot summer afternoon it suddenly rains and cools down. This quick change is an example of:
- (A) the Sun cooling down
 - (B) weather changing
 - (C) the season ending
 - (D) climate changing

78. If one angle of two intersecting lines is 120° , the angle next to it (linear pair) is:
- (A) 120°
 - (B) 240°
 - (C) 90°
 - (D) 60°
79. Which is the odd one out (NOT a chemical change)?
- (A) melting of butter
 - (B) rusting of iron
 - (C) cooking of food
 - (D) burning of paper
80. Why do desert animals like the camel and desert fox often have large feet or big ears?
- (A) to cope with sand and heat (big feet for sand, big ears to lose heat)
 - (B) to fly
 - (C) to store water in ears
 - (D) to swim in rivers
81. An angle and its vertically opposite angle together with the two other angles around the point add up to:
- (A) 360°
 - (B) 270°
 - (C) 90°
 - (D) 180°
82. Which process forms a NEW substance and so is a chemical change?
- (A) chilling water
 - (B) tearing a roti
 - (C) grinding wheat
 - (D) digestion of food in our body
83. Tropical rainforests have a climate that is:
- (A) cold and wet
 - (B) cold and dry
 - (C) hot and dry
 - (D) hot and wet with rain through the year
84. Two angles are supplementary and equal. Each is:
- (A) 180°
 - (B) 90°
 - (C) 45°
 - (D) 60°

85. Which of these everyday changes is reversible (physical)?
- (A) a firework exploding
 - (B) milk turning into curd
 - (C) an egg being fried
 - (D) ice cream melting on a hot day
86. Which instrument measures wind speed?
- (A) hygrometer
 - (B) thermometer
 - (C) rain gauge
 - (D) anemometer
87. If one angle of a linear pair is a right angle (90°), the other angle is:
- (A) 45°
 - (B) 180°
 - (C) 0°
 - (D) 90°
88. Forming iron sulphide by heating iron and sulphur is chemical because the product:
- (A) is still a simple mixture
 - (B) can be split easily
 - (C) is colder than before
 - (D) has new properties and can't be separated by a magnet
89. Which factor most directly decides whether you carry an umbrella TODAY?
- (A) last year's rainfall
 - (B) the region's climate
 - (C) today's weather forecast
 - (D) the country's flag
90. The three angles of a triangle (a related fact) always add up to:
- (A) 90°
 - (B) 180°
 - (C) 270°
 - (D) 360°
91. Which of these is a chemical change?
- (A) melting butter
 - (B) breaking chalk
 - (C) rusting of an iron gate
 - (D) boiling water
92. A hygrometer is the instrument used to measure:
- (A) rainfall
 - (B) temperature
 - (C) wind speed
 - (D) humidity (water vapour in air)

93. Two equal angles are complementary. Each angle is:
- (A) 45°
 - (B) 90°
 - (C) 30°
 - (D) 60°
94. Tearing a sheet of paper into pieces is a:
- (A) irreversible chemical change
 - (B) physical change
 - (C) burning
 - (D) chemical change
95. A penguin's smooth, streamlined body mainly helps it to:
- (A) climb tall trees
 - (B) swim swiftly through water to catch fish
 - (C) dig deep burrows
 - (D) fly long distances
96. The angle between the two hands of a clock at exactly 3 o'clock is:
- (A) 180°
 - (B) 90° (a right angle)
 - (C) 45°
 - (D) 60°
97. A cut apple slowly turning brown in air shows a chemical change because:
- (A) a new brown substance forms (and it can't be undone)
 - (B) it only changes shape
 - (C) it just gets colder
 - (D) it can be turned white again
98. Many rainforest frogs are brightly coloured mainly to:
- (A) swim faster
 - (B) warn predators that they may be poisonous
 - (C) store water
 - (D) keep themselves warm
99. An angle that measures 200° is a:
- (A) straight angle
 - (B) right angle
 - (C) obtuse angle
 - (D) reflex angle
100. The surest way to decide if a change is chemical (not physical) is to check whether:
- (A) the object got wet
 - (B) the object moved
 - (C) the object changed shape
 - (D) a new substance with new properties has formed